

Photoelectric Effect

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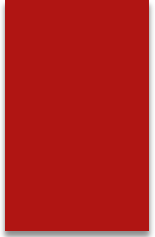
Wave Nature of Light:

Christian Huygens proposed that the light is propagated in the form of a wave. It was not able to explain the propagation of light in a vacuum. This drawback was removed by Maxwell he proposed that light is an electromagnetic wave and for the propagation of electromagnetic waves no material medium is required. Thus the wave nature of light was established.

Wave theory was able to explain all the phenomena associated with the propagation of light. But it failed to explain the energy distribution and modern phenomenon like Photoelectric effect, Compton effect, etc.

Particle Nature of Light:

Max Planck proved that the propagation of light or energy takes place in the form of packets of energy called quanta. Quantum of light is called a photon and thus he established the particle nature of light. Using particle or quantum nature of radiation we can explain the phenomenon of photoelectric effect and Compton effect.



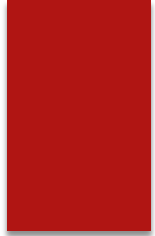
Planck's Quantum Theory:

The quantum theory was proposed by Max Planck. According to this theory, radiation from a source is not emitted continuously, but it is emitted in packets or bundles of energy. These packets are called quanta or photons. If the radiation is of frequency ν , each quantum has energy where h is Planck's constant.

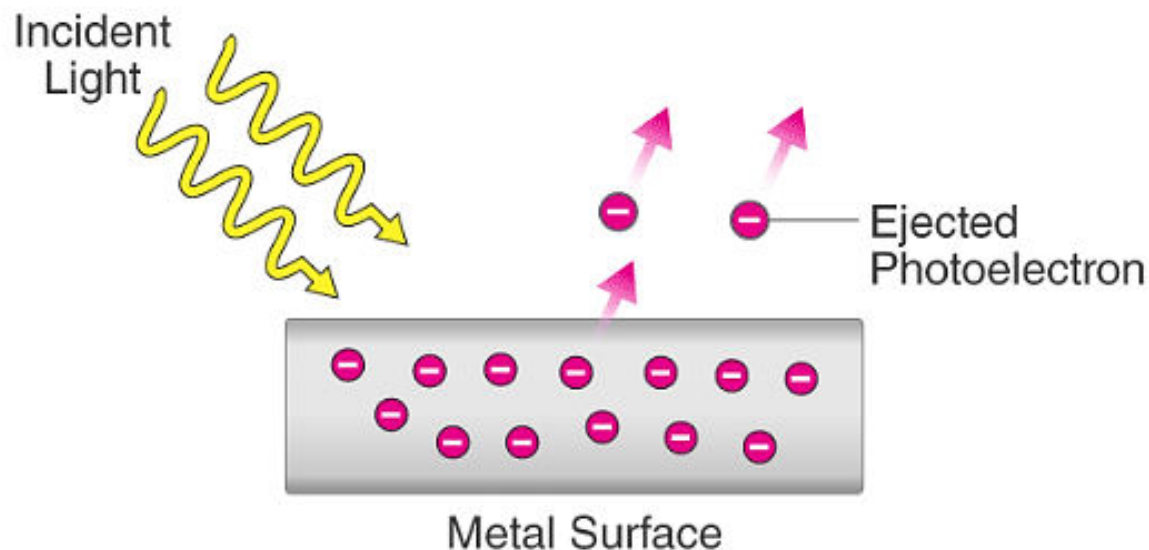
Thus energy of photon = $E = h\nu$

The photoelectric effect

The **photoelectric effect** is a phenomenon in which electrons are ejected from the surface of a metal when light is incident on it. These ejected electrons are called **photoelectrons**. It is important to note that the emission of photoelectrons and the kinetic energy of the ejected photoelectrons is dependent on the frequency of the light that is incident on the metal's surface. The process through which photoelectrons are ejected from the surface of the metal due to the action of light is commonly referred to as **photoemission**.



The photoelectric effect occurs because the electrons at the surface of the metal tend to absorb energy from the incident light and use it to overcome the attractive forces that bind them to the metallic nuclei. An illustration detailing the emission of photoelectrons as a result of the photoelectric effect is provided below.



Explaining the Photoelectric Effect: The Concept of Photons

The photoelectric effect **cannot** be explained by considering light as a **wave**. However, this phenomenon **can** be explained by the **particle** nature of light, in which light can be visualized as a stream of packets/particles of electromagnetic energy. These 'particles' of light are called **photons**. The energy held by a photon is related to the frequency of the light via [Planck's equation](#).

$$E = h\nu = hc/\lambda$$

Where,

- E denotes the energy of the photon
- h is Planck's constant
- ν denotes the frequency of the light
- c is the speed of light (in a vacuum)
- λ is the wavelength of the light

Thus, it can be understood that **different frequencies of light carry photons of varying energies**. For example, the frequency of blue light is greater than that of red light (the wavelength of blue light is much shorter than the wavelength of red light). Therefore, the energy held by a photon of blue light will be greater than the energy held by a photon of red light.

Threshold energy/work function

For the photoelectric effect to occur, the photons that are incident on the surface of the metal must carry sufficient energy to overcome the attractive forces that bind the electrons to the nuclei of the metals. The minimum amount of energy required to remove an electron from the metal is called the **threshold energy** (denoted by the symbol Φ).

For a photon to possess energy equal to the threshold energy, its frequency must be equal to the **threshold frequency** (which is the minimum frequency of light required for the photoelectric effect to occur). The threshold frequency is usually denoted by the symbol ν_{th} , and the associated wavelength (called the threshold wavelength) is denoted by the symbol λ_{th} .

The relationship between the threshold energy and the threshold frequency can be expressed as follows.

$$\Phi = h\nu_{\text{th}} = hc/\lambda_{\text{th}}$$

Relationship between the Frequency of the Incident Photon and the Kinetic Energy of the Emitted Photoelectron

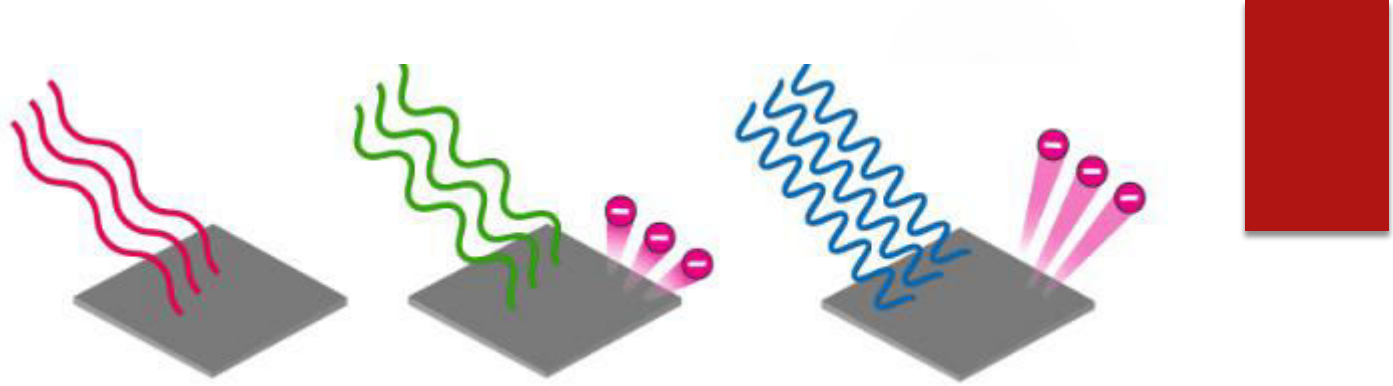
The relationship between the energy of the photon and the kinetic energy of the emitted photoelectron can be written as follows:

$$E_{\text{photon}} = \Phi + E_{\text{electron}} \\ \Rightarrow h\nu = h\nu_{\text{th}} + \frac{1}{2}m_e v^2$$

Where,

- E_{photon} denotes the energy of the incident photon, which is equal to $h\nu$
- Φ denotes the threshold energy of the metal surface, which is equal to $h\nu_{\text{th}}$
- E_{electron} denotes the kinetic energy of the photoelectron, which is equal to $\frac{1}{2}m_e v^2$ (m_e = Mass of electron = 9.1×10^{-31} kg)

If the energy of the photon is less than the threshold energy, there will be no emission of photoelectrons (since the attractive forces between the nuclei and the electrons cannot be overcome). Thus, the photoelectric effect will not occur if $\nu < \nu_{\text{th}}$. If the frequency of the photon is exactly equal to the threshold frequency ($\nu = \nu_{\text{th}}$), there will be an emission of photoelectrons, but their kinetic energy will be equal to zero. An illustration detailing the effect of the frequency of the incident light on the kinetic energy of the photoelectron is provided below.



From the image, it can be observed that

- The photoelectric effect does not occur when the red light strikes the metallic surface because the frequency of red light is lower than the threshold frequency of the metal.
- The photoelectric effect occurs when green light strikes the metallic surface, and photoelectrons are emitted.
- The photoelectric effect also occurs when blue light strikes the metallic surface. However, the kinetic energies of the emitted photoelectrons are much higher for blue light than for green light. This is because blue light has a greater frequency than green light.

Why the threshold energy varies from metal to metal?

It is important to note that the threshold energy varies from metal to metal. This is because the attractive forces that bind the electrons to the metal are different for different metals. It can also be noted that the photoelectric effect can also take place in non-metals, but the threshold frequencies of non-metallic substances are usually very high.

Table 1 lists some values of the work function of different materials.

Table 1: Some photoelectric work-functions

Material	Na	Al	Co	Cu	Zn	Ag	Pt	Pb
Workfunction $\Phi(\text{eV})$	2.28	4.08	3.90	4.70	4.31	4.73	6.35	4.14

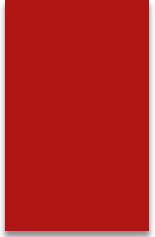
Einstein's Photoelectric Equation:

On the basis of Planck's quantum theory, Einstein derived an equation for the photoelectric effect known as **Einstein photoelectric equation**. For this excellent work, he was honored with the Nobel Prize in 1921. Einstein assumed that

- Light consists of photons or quanta of energy, energy in each photon is $h\nu$. Where h is Planck's constant and ν is the frequency of light
- Each incident photon collides with an electron inside an atom and gives all its energy to the electron.
- Part of this energy is used by the electron to come out of the surface of the metal and the remaining part is the kinetic energy with which the electron is emitted.
- The minimum energy required by an electron to come out of the surface of the metal is called the photoelectric work function (ϕ) of the metal.
- The remaining energy ($h\nu - \phi$) is the maximum kinetic energy of the electron with which a photoelectron will be ejected.

Thus, Maximum kinetic energy of electron = energy of photon – work function

Let ' m ' be the mass of electron and $v_{\max}$ be the maximum velocity of photo-electron by which it will be ejected.


$$\therefore \frac{1}{2} mv_{\max}^2 = h\nu - \phi$$

$$\therefore h\nu = \phi + \frac{1}{2} mv_{\max}^2$$

$$\text{But } \phi = h\nu_0$$

Where ν_0 = Threshold frequency.

$$\therefore h\nu = h\nu_0 + \frac{1}{2} mv_{\max}^2$$

$$\therefore \frac{1}{2} mv_{\max}^2 = h\nu - h\nu_0$$

$$\therefore \frac{1}{2} mv_{\max}^2 = h(\nu - \nu_0)$$

This equation is known as Einstein's photoelectric equation

Conversion of KE from joules to eV

For a particular metal

$$K_{max} = h(\nu - \nu_c) = h\left(\frac{c}{\lambda} - \frac{c}{\lambda_c}\right) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_c}\right) = 6.625 \times 10^{-34} \times 3 \times 10^8 \left(\frac{1}{\lambda} - \frac{1}{\lambda_c}\right) \text{ joules, } \lambda \text{ and } \lambda_c \text{ in meter}$$
$$= \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{10^{-10} \times 1.602 \times 10^{-19}} \left(\frac{1}{\lambda} - \frac{1}{\lambda_c}\right) \text{ eV, } \lambda \text{ and } \lambda_c \text{ in Angstrom} = 12400 \left(\frac{1}{\lambda} - \frac{1}{\lambda_c}\right) \text{ eV, } \lambda \text{ and } \lambda_c \text{ in Angstrom}$$

Planck's the constant value is $6.62507015 \times 10^{-34}$ joule-hertz⁻¹ (or joule-seconds).

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

1 electron volt is the energy change that takes place when a charge equal to 1 electron ($1.6 \times 10^{-19} \text{ C}$) is moved through a potential difference of 1 volt.

Laws Governing the Photoelectric Effect

- For a light of any given frequency,; ($\gamma > \gamma_{Th}$), the photoelectric current is directly proportional to the intensity of light.
- For any given material, there is a certain minimum (energy) frequency, called threshold frequency, below which the emission of photoelectrons stops completely, no matter how high the intensity of incident light is.
- The maximum kinetic energy of the photoelectrons is found to increase with the increase in the frequency of incident light, provided the frequency ($\gamma > \gamma_{Th}$) exceeds the threshold limit.
- The maximum kinetic energy is independent of the intensity of light.
- The photo-emission is an instantaneous process.

Photo-electric work function (Joules to eV)

$$\phi = h \nu_c = \frac{hc}{\lambda_c} = 6.625 \times 10^{-34} \times 3 \times 10^8 \frac{1}{\lambda_c} \text{ joules, } \lambda_c \text{ in meter}$$

$$= \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{10^{-10} \times 1.602 \times 10^{-19}} \frac{1}{\lambda_c} \text{ eV, } \lambda_c \text{ in Angstrom}$$

$$= \frac{12400}{\lambda_c} \text{ eV, } \lambda_c \text{ in Angstrom}$$

(10)

Problems of Photo-electric effect

Example 1: (a) what are the energy and momentum of a photon of red light of wavelength 650 nm? (b) What is the wavelength of a photon of energy 2.40 eV?

Solution:

(a)

$$E = h\nu = \frac{hc}{\lambda} = \frac{[6.63 \times 10^{-34} \text{ J.s}][3.0 \times 10^8 \text{ m/s}]}{650 \times 10^{-9} \text{ m}} = 3.06 \times 10^{-19} \text{ J} = \frac{3.06 \times 10^{-19} \text{ J}}{1.60 \times 10^{-19} \text{ J/eV}} = 1.91 \text{ eV}$$

(b)

$$\lambda = \frac{hc}{E} = \frac{1240 \text{ eV.nm}}{2.40 \text{ eV}} = 517 \text{ nm}$$

Example 2: Calculate the threshold frequency and the corresponding wavelength of radiation incident on a certain metal whose work function is 3.31×10^{-19} joules. Given, Planck's constant = 6.62×10^{-34} Joule-sec.

Solution:

• Work function, $\phi = h\nu_c \Rightarrow \nu_c = \frac{\phi}{h} \Rightarrow \nu_c = \frac{3.31 \times 10^{-19} \text{ J}}{6.625 \times 10^{-34} \text{ J.s}} = 5 \times 10^{14} \text{ Hertz}$

$$\lambda_c = \frac{c}{\nu_c} = \frac{3 \times 10^8 \text{ m/sec}}{5 \times 10^{14} / \text{sec}} = 6 \times 10^{-7} \text{ m} = 6000 \text{ Angstrom}$$

Example 3: The work-function for tungsten metal is 4.52 eV. (a) What is the cut-off wavelength λ_c for tungsten? (b) What is the maximum K. E of the electron when radiation of wavelength 198 nm is used?

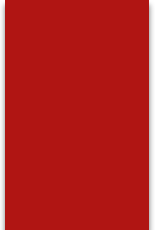
Solution:

(a) We know that the work function can be written as $\phi = h\nu_c = \frac{hc}{\lambda_c}$

Therefore, we can write the cutoff wavelength $\lambda_c = \frac{hc}{\phi} = \frac{1240 \text{ eV} \cdot \text{nm}}{4.52 \text{ eV}} = 274 \text{ nm}$

(b) Maximum kinetic energy of the photoelectron can be written as

$$K.E_{\text{max}} = h\nu - h\nu_c = \frac{hc}{\lambda} - \phi = \frac{1240 \text{ eV} \cdot \text{nm}}{198 \text{ nm}} - 4.52 \text{ eV} = 1.74 \text{ eV}$$



Example 4: Light of wavelength 4300 Angstrom is incident on (a) nickel surface of work function 5 eV and (b) potassium surface of work function 2.3 eV. Find out, if electrons will be emitted, and if so, the maximum velocity of the emitted electrons in each cases.

Solution:

Here

Light wavelength, $\lambda = 4300$ Angstrom

Work function for Nickel, $\phi_{Ni} = 5 \text{ eV}$

Work function for Potassium, $\phi_P = 2.3 \text{ eV}$

(i) For the nickel surface

We know that work function, $\phi_{Ni} = h\nu_c = \frac{hc}{\lambda_c} = \frac{12400}{\lambda_c} \text{ eV} \Rightarrow \lambda_c = \frac{12400}{\phi_{Ni}} = \frac{12400}{5} \text{ \AA} = 2480 \text{ \AA}$

Since $\lambda_c < \lambda$, no electrons will be emitted from the Nickel surface.

(ii) For the Potassium surface

Work function, $\phi_P = \frac{12400}{\lambda_c} \text{ eV} \Rightarrow \lambda_c = \frac{12400}{\phi_P} = \frac{12400}{2.3} \text{ \AA} = 5391 \text{ \AA}$

Since $\lambda_c > \lambda$, electrons will be emitted from the Potassium surface

Maximum kinetic energy of photo-electrons is

$$K_{\max} = 12400 \left(\frac{1}{\lambda} - \frac{1}{\lambda_c} \right) \text{ eV, } \lambda \text{ and } \lambda_c \text{ in Angstrom}$$

$$= 12400 \left(\frac{1}{4300} - \frac{1}{5391} \right) \text{ eV} = 12400 \left(\frac{5391 - 4300}{4300 \times 5391} \right) \text{ eV} = 0.583 \text{ eV}$$

$$\text{Or, } \frac{1}{2} m v_{\max}^2 = 0.583 \text{ eV} = 0.583 \times 1.602 \times 10^{-19} \text{ J} = 0.9349 \times 10^{-19} \text{ J}$$

$$\Rightarrow v_{\max} = \sqrt{\frac{2 \times 0.9349 \times 10^{-19} \text{ J}}{m}} = \sqrt{\frac{1.8698 \times 10^{-19} \text{ J}}{9.1 \times 10^{-31} \text{ kg}}} = \sqrt{0.2054 \times 10^{12}} = 4.6 \times 10^5 \text{ m/s}$$

Example 5: The photoelectric threshold of copper is 3200 Angstrom. If ultra-violet light of wavelength 2500 Angstrom falls on it, find (a) the maximum kinetic energy of the photo-electrons ejected, (b) maximum velocity of the photo-electrons and (c) the value of the work function.

(a) Maximum kinetic energy of photo-electrons is

$$K_{\max} = 12400 \left(\frac{1}{\lambda} - \frac{1}{\lambda_c} \right) \text{ eV, } \lambda \text{ and } \lambda_c \text{ in Angstrom}$$
$$= 12400 \left(\frac{1}{2500} - \frac{1}{3200} \right) \text{ eV} = 1.087 \text{ eV}$$

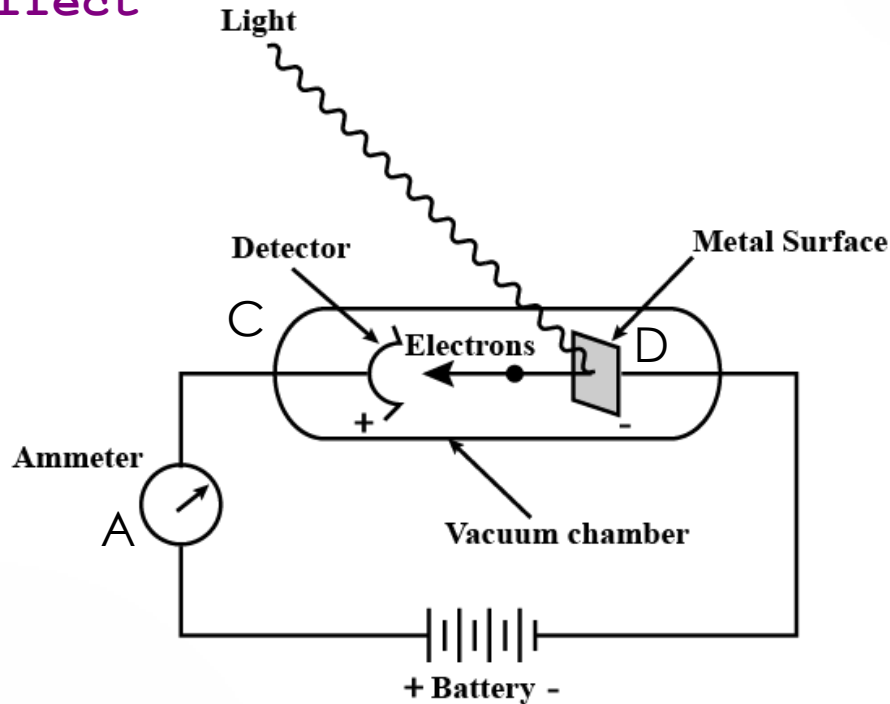
(b) Maximum velocity of the photo-electrons is given by

$$K_{\max} = \frac{1}{2} m v_{\max}^2 = 1.087 \text{ eV} = 1.087 \times 1.602 \times 10^{-19} \text{ J}$$
$$\Rightarrow v_{\max} = \sqrt{\frac{2 \times 1.087 \times 1.602 \times 10^{-19} \text{ J}}{m}} = \sqrt{\frac{3.482748 \times 10^{-19} \text{ J}}{9.1 \times 10^{-31} \text{ kg}}} = 6.186 \times 10^5 \text{ m/sec}$$

$$(c) \text{ Work function } \phi = h \nu_c = \frac{hc}{\lambda_c} = \frac{12400}{\lambda_c} \text{ eV, } \lambda_c \text{ in A.U.} = \frac{12400}{3200} \text{ eV} = 3.88 \text{ eV}$$

Experimental Study of the Photoelectric Effect

L3



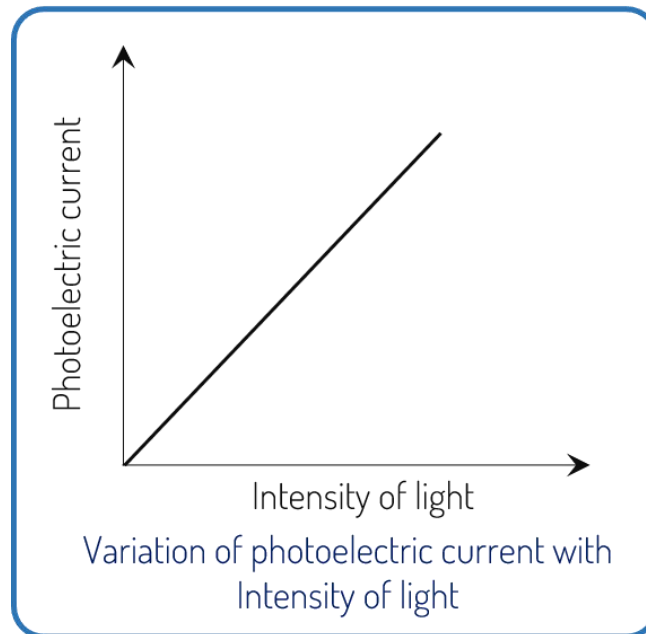
The given experiment is used to study the photoelectric effect experimentally. In an evacuated glass tube, two zinc plates, C and D, are enclosed. Plate C acts as an anode, and D acts as a photosensitive plate. Two plates are connected to battery B and ammeter A. If the radiation is incident on plate D through a quartz window, electrons are ejected out of the plate, and current flows in the circuit. This is known as photocurrent. Plate C can be maintained at desired potential (+ve or – ve) with respect to plate D.

Effects of Intensity of Incident Radiation on Photoelectric Effect

The potential difference between the metal plate, collector and frequency of incident light is kept constant, and the intensity of light is varied.

The electrode C, i.e., the collecting electrode, is made positive with respect to D (metal plate). For a fixed value of frequency and the potential between the metal plate and collector, the photoelectric current is noted in accordance with the intensity of incident radiation.

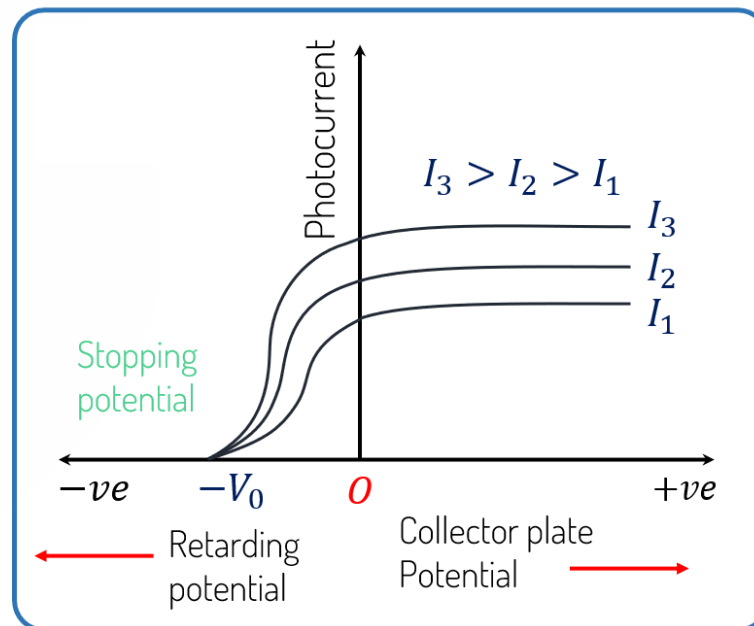
It shows that photoelectric current and intensity of incident radiation both are proportional to each other. The photoelectric current gives an account of the number of photoelectrons ejected per sec.



Effects of Potential Difference between the Metal Plate and the Collector on the Photoelectric Effect

The frequency of incident light and intensity is kept constant, and the potential difference between the plates is varied.

The collector plate anode is initially kept positive with respect to the emitter plate. As the potential at the anode is increased, the photocurrent measured by the ammeter also increases. Even at zero potential, some photocurrent exists due to the already emitted electrons which move because of their kinetic energy. The photocurrent increases up to a certain level after which it becomes saturated since the number of electrons emitted remains constant. When the polarity is reversed, the photocurrent starts decreasing and becomes zero at a particular value of retarding potential known as stopping potential.

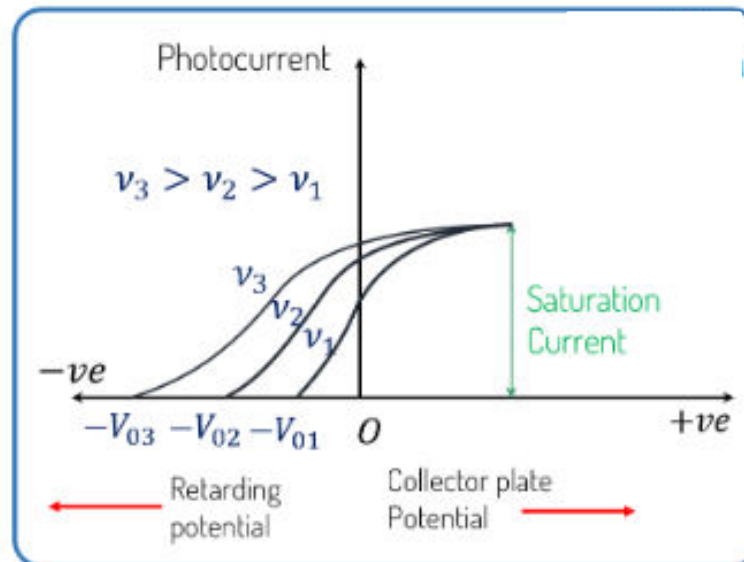


Effect of Frequency on Photoelectric Effect

The intensity of light is kept constant, and the frequency of light is varied.

For a fixed intensity of incident light, variation in the frequency of incident light produces a linear variation of the cut-off potential/stopping potential of the metal. It is shown that the cut-off potential (V_c) is linearly proportional to the frequency of incident light.

The kinetic energy of the photoelectrons increases directly proportionally to the frequency of incident light to completely stop the photoelectrons. We should reverse and increase the potential between the metal plate and collector in (negative value) so the emitted photoelectron can't reach the collector.



The intensity of incident radiation and the kinetic energy of photoelectrons

Typical experimental curves are shown in Figure 6.3.2, in which the photocurrent is plotted versus the applied potential difference between the electrodes. For the positive potential difference, the current steadily grows until it reaches a plateau. Furthering the potential increase beyond this point does not increase the photocurrent at all. A higher intensity of radiation produces a higher value of photocurrent. For the negative potential difference, as the absolute value of the potential difference increases, the value of the photocurrent decreases and becomes zero at the stopping potential. For any intensity of incident radiation, whether the intensity is high or low, the value of the stopping potential always stays at one value.

To understand why this result is unusual from the point of view of classical physics, we first have to analyze the energy of photoelectrons. A photoelectron that leaves the surface has kinetic energy K . It gained this energy from the incident electromagnetic wave. In the space between the electrodes, a photoelectron moves in the electric potential and its energy changes by the amount $q\Delta V$, where ΔV is the potential difference and $q = -e$. Because no forces are present but electric force, by applying the work-energy theorem, we obtain the energy balance $\Delta K - e\Delta V = 0$ for the photoelectron, where ΔK is the change in the photoelectron's kinetic energy. When the stopping potential $-\Delta V_s$ is applied, the photoelectron loses its initial kinetic energy K_i and comes to rest. Thus, its energy balance becomes $(0 - K_i) - e(-\Delta V_s) = 0$, so that $K_i = e\Delta V_s$. In the presence of the stopping potential, the largest kinetic energy K_{max} that a photoelectron can have is its initial kinetic energy, which it has at the surface of the photoelectrode. Therefore, the largest kinetic energy of photoelectrons can be directly measured by measuring the stopping potential:

$$K_{max} = e\Delta V_s. \tag{6.3.1}$$

At this point we can see where the classical theory is at odds with the experimental results. In classical theory, the photoelectron absorbs electromagnetic energy in a continuous way; this means that when the incident radiation has a high intensity, the kinetic energy in Equation 6.3.1 is expected to be high. Similarly, when the radiation has a low intensity, the kinetic energy is expected to be low. But the experiment shows that the maximum kinetic energy of photoelectrons is independent of the light intensity.

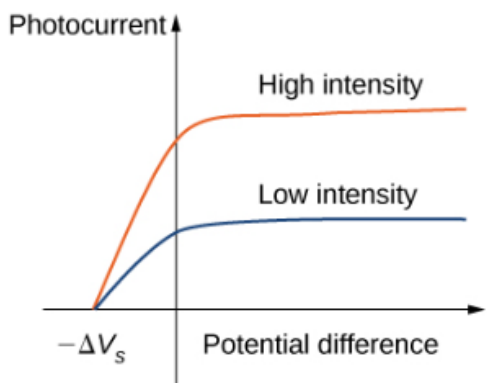
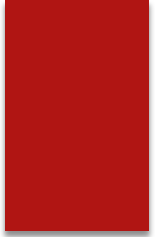


Figure 6.3.2: The detected photocurrent plotted versus the applied potential difference shows that for any intensity of incident radiation, whether the intensity is high (upper curve) or low (lower curve), the value of the stopping potential is always the same.



Applications of the Photoelectric Effect

- Used to generate electricity in solar panels. These panels contain metal combinations that allow electricity generation from a wide range of wavelengths.
- Lighting sensors, such as the ones used in smartphones, enable automatic adjustment of screen brightness according to the lighting. This is because the amount of current generated via the photoelectric effect is dependent on the intensity of light hitting the sensor.
- Digital cameras can detect and record light because they have photoelectric sensors that respond to different colours of light.
- X-Ray Photoelectron Spectroscopy (XPS):** This technique uses X-rays to irradiate a surface and measure the kinetic energies of the emitted electrons. Important aspects of the chemistry of a surface can be obtained, such as elemental composition, chemical composition and chemical state.